



RUBEN PEREZ

+1 408-430-4740 • perezben37@gmail.com • [linkedin.com/in/rubaduben](https://www.linkedin.com/in/rubaduben)

Website:



SUMMARY

Enthusiastic and results-driven Computer Engineer with a strong robotics and software engineering background. I am seeking to leverage my expertise in autonomous systems and software development to contribute to innovative projects in the field of robotics.

EDUCATION

Bachelor of Science | *Computer Engineering with a Concentration in Robotics* September 2017 – June 2021
University of California, Santa Cruz Santa Cruz, CA

PROJECTS

Custom OBD2 Python Module | *Python* January 2024 – February 2024

Independent

- Developed an *asyncio* and *SocketCAN*-based Python module for interfacing with vehicle's OBD2 systems. The module features a *CANPort* object for socket management, raw CAN datagram packet handling, and CAN port filter configuration.

STM32 Wireless CAN Transceiver/Receiver | *C/C++* March 2024 - Present

Independent

- Developing firmware for an embedded system using an *STM32* microcontroller to interface with a vehicle's OBD2 port and transmit CAN packets wirelessly using an nRF module.

WORK EXPERIENCE

Advanced Farm Davis, CA

Software Engineering Manager of Testing October 2022 – Present

- Lead software development efforts at an agricultural robotics startup, specializing in autonomous fruit harvesting using machine learning techniques.
- Successfully managed a cross-functional team of 3 software test engineers and 3 software support engineers
- Designed a network configuration (*Ethernet, TCP/IP*) for a distributed compute architecture, enabling field hardware replacement without reprogramming and reducing downtime.
- Led a project to migrate our robot's compute and vision stack from a combination of an *Intel NUC* and *Nvidia Jetson TX2* with a proprietary camera to a single *Nvidia Jetson Orin AGX* computer with a mini *Zed* camera.
- Designed and implemented a pipeline for receiving, extracting, and storing technical data from an automated packing system. (*Debian, Prometheus, SQL, Grafana, Python*).

↳ *Software Test Engineer* November 2021 – October 2022

- Commissioned and validated a fleet of 16 robotic strawberry harvesters, ensuring the successful integration and functionality of their sub-assemblies, firmware, and individual components.
- Optimized and took ownership of a critical robotic sub-system within three months, Effectively reducing position following error by a factor of 5 and greatly improved driveability in rough terrain. *Python, Twisted*, and *ZeroMQ*.
- Developed command line tool (Python) to decode CAN traffic into *CANopen* messages and plot-relevant sensor data and device state information.

↳ *Software Support Engineer* March 2021 – November 2021

- Monitored a fleet of strawberry harvesters in the field, providing software support and conducting maintenance to optimize performance and minimize downtime.

SKILLS

Languages: English (Fluent), Spanish (Native)

Programming: Python (NumPy, SciPy, Matplotlib, Twisted, ZMQ, Seaborn, Pandas, Numba, Asyncio), MATLAB, C/C++, Bash, Git, HTTP

Software: ROS, Github, Linux, Gazebo, Excel, Fusion 360, stm32 CubeIDE